AQUEDUCTx

A generalized peer-to-peer logistics and finance layer
for smallholder farmers.

aqueductx.trade · open-source · MIT

Judgment doesn't scale down.

A third of the world's food is grown by smallholders. Between them and the markets that consume it stands a chain of intermediaries, because trade runs on judgment: someone must grade the goods, vouch for their origin, find the buyer, and carry the risk. That judgment is an operational cost a single smallholder lot cannot carry. And lenders do not lend against lots they cannot verify—**legibility comes before capital.**

$2.5T

The global trade-finance gap—credit demanded worldwide that goes unmet.

ADB Trade Finance Gaps Survey, 2023 & Jan 2026

$250

The minimum inspection fee per shipment—a smallholder lot cannot carry it.

Bureau Veritas Verigates PVoC fee schedule

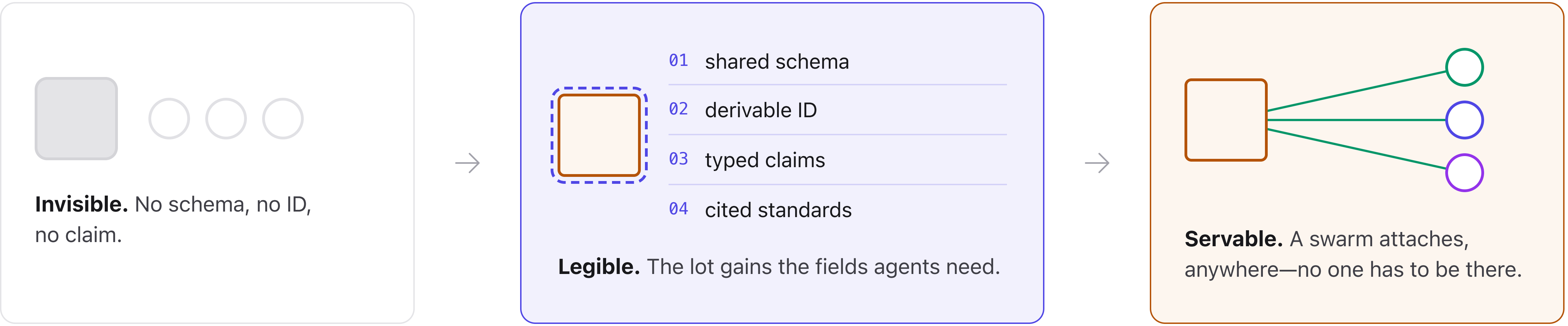
$323B

Annual smallholder finance demand, against roughly $95B supplied.

ISF Advisors Global Report, 2025

A legible lot is a servable lot.

A smallholder's lot fails three checks lenders require: it cannot afford professional verification, it must re-prove compliance to every new buyer, and nobody can confirm the harvest is not pledged twice. Make the lot legible—a shared schema, a derivable ID, typed claims—and agents run all three checks at almost no cost: one proof serves every standard, double-pledging becomes a registry lookup, and verification works at single-lot scale.



Sources: Bureau Veritas Verigates PVoC fee schedule ($250 inspection floor) · MonetaGo duplicate-financing registry, live since 2018 · EU Deforestation Regulation (EUDR), in force.

The producer gets a seat, not a terminal.

Automated judgment placed in the producer's hands: a steward agent that represents producer interests and nothing else—extensible with negotiation strategies, alerts, or any automation that serves them.

THE SEAT

Steward agent

bound to one coop identity

Holds—the producer's plot geometry · credit history · reserve floor · harvest windows · disclosure preferences

Does—screens solver bids against the floor, assembles compliance evidence once, prepares settlement for confirmation

FIVE VERBS, ONE SURFACE

01 **Post an intent**
sell a lot, finance a planting, source an input.

02 **Set the floor**
the producer's reserve price; solvers bid against it, never under it.

03 **Accept or decline a fill**
matches arrive as named terms; the producer decides.

04 **Confirm settle**
human-confirmed by default—fully delegable to the steward by designation

05 **Set disclosure tiers**
decides what evidence leaves the seat; plot geometry stays.

One open layer, a swarm of agents.

Single-purpose agents act **concurrently** on a small set of shared objects. No agent owns the next; each reads and writes the same lots, intents, claims, and standards.

Scout
aggregates real platform surfaces

Diligence
verifies origin & grade, flags gaps

Certifier-resolver
resolves cited standards: TIC · GLOW-GCA · GIIN

→ ←

SHARED OBJECTS · EVERY AGENT READS AND WRITES THESE

Lots
the resource: commodity, grade, origin

Intents
sell-this-lot · finance-this-farm

Claims
financing's deferred half

Standards
cited conditions on routes & bids

→ ←

Oracle
prices against a real floor

Solver
races to fill, landed-cost bids

Steward
settles & custodies for the coop

ONE LOT'S LIFECYCLE verify → price → settle

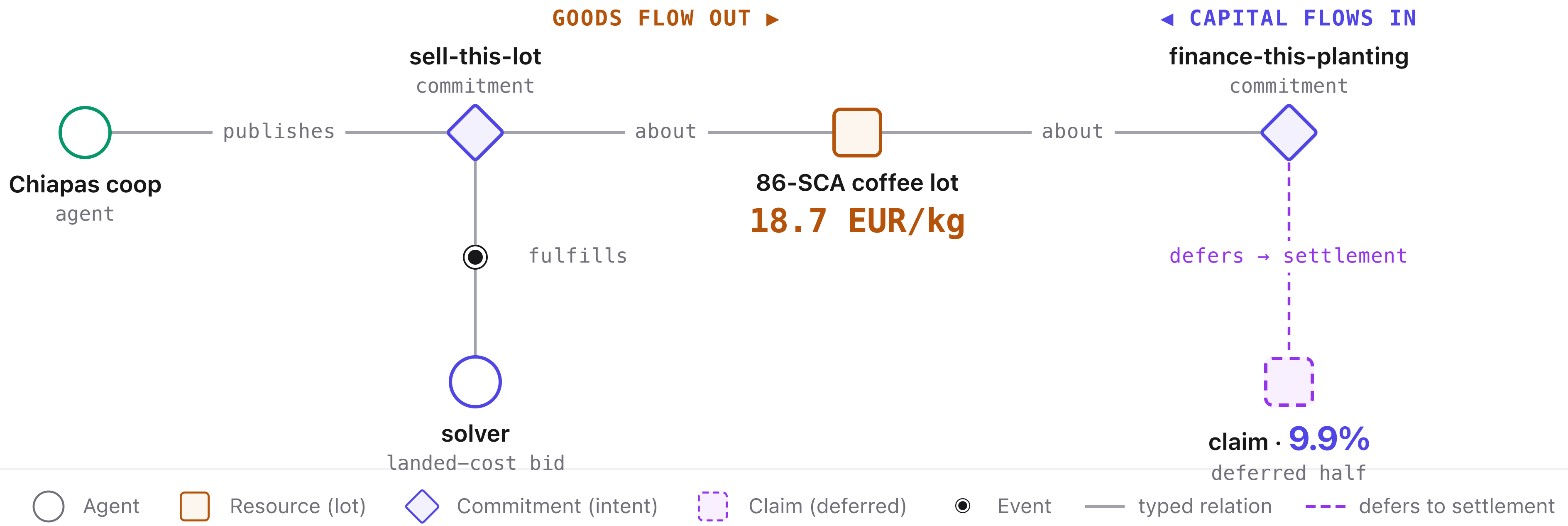
The data model is REA—the Claim is the point.

Selling a harvest and financing one are the same kind of exchange. AqueductX models the smallholder economy as **Resources, Events, and Agents**—what a farm sells and what it consumes to produce, in one schema. Writing the model down exposed the one object every platform lacks: the **Claim**. Capital moves now; the repayment owed later is a first-class record.

REA CONCEPT	AQUEDUCTX TYPE	EXAMPLE
Resource	`AqueductAnyLot` `+ inputResource`	**Output**—a lot the farm sells: coffee, weight, grade. **Input**—what the farm consumes: seedlings, home solar (typed `inputResource`).
Agent	`AqueductActor + hubs`	cooperative, venue, solver, certifier, infrastructure
Commitment	`AqueductIntent`	sell-this-lot · finance-this-planting · finance-this-farm
Event	`AqueductEvent`	a fill—merged with real per-event source URLs
Claim	`claim: {principal, aprPct, term}`	**A Claim is a repayment owed later for capital advanced now.** The rate shown is real: published by a live lending platform, with completed repayment cycles onchain. `Source: EthicHub / Heifer facility documentation.`

Coffee—the loop, running on real EthicHub reads.

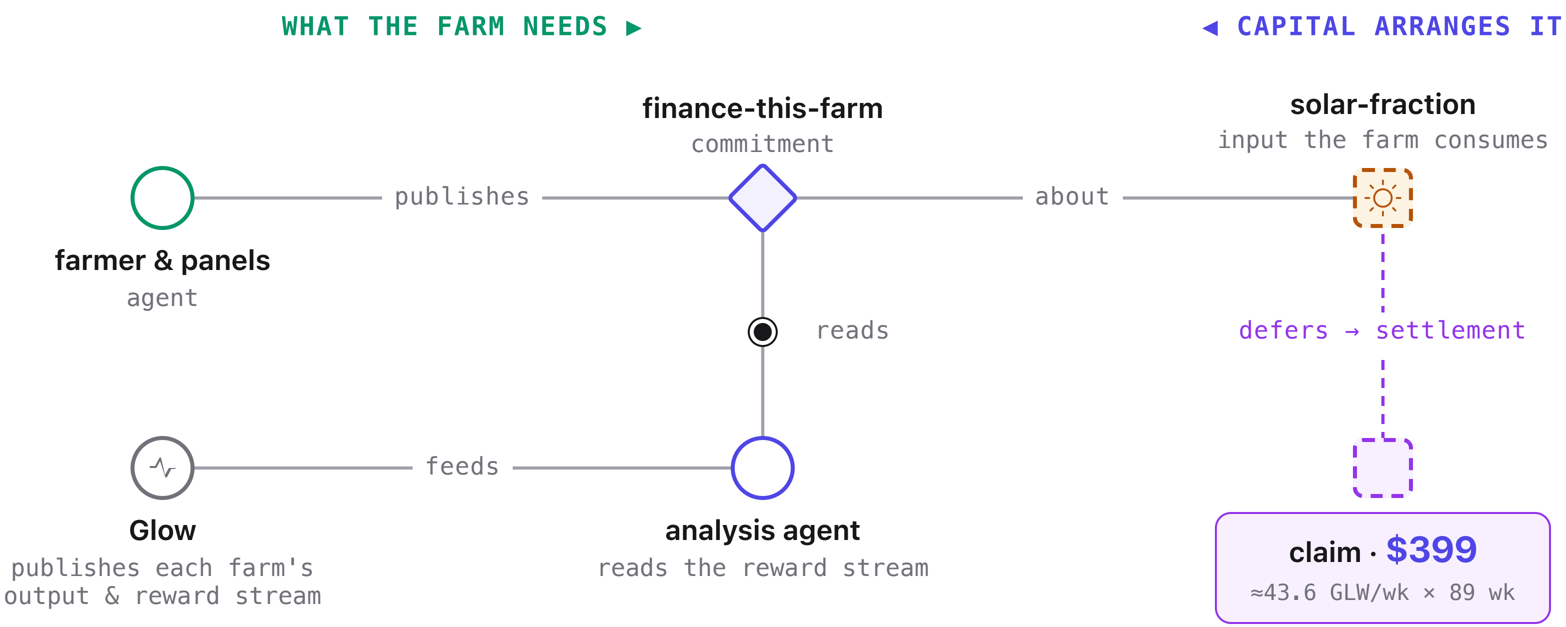
A real Chiapas lot, read live from EthicHub, a crowdlending platform financing smallholder coffee cooperatives. The coop publishes a sell-intent; a solver's fill fulfills it. Beside it, a finance-intent carries a Claim—capital now, repayment at the platform's published **9.9%** —the deferred half of the same exchange.



Live read: greencoffee.ethichub.com · rate: EthicHub/Heifer facility

The same swarm that sells your coffee finances your panels.

A farmer needs energy to produce. Glow, a protocol financing distributed solar, publishes each farm's data; an analysis agent finds the financeable input and arranges the terms—the farmer sees the panel and the repayment.



124 audited farms read live · live GLW **≈$0.2825** from the onchain pool—the agent reads a real stream, not a projection. Source: glow.org public audits API · GLW from the onchain pool.

Two builders who ship, with institutional reach.

Patrick Rawson

Co-Founder & CEO

Full-stack builder across AI orchestration and blockchain mechanism design; ex-CEO of Curve Labs. Co-author (Taylor & Francis), *The Green Crypto Handbook: Blockchain for Sustainability Professionals*.

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Louise Borreani

Co-Founder

Researcher and strategist across AI, blockchain, and sustainability (MA Environmental Policy, Sciences Po Paris); authored the dMRV research underpinning Regen Atlas. Co-author, *The Green Crypto Handbook: Blockchain for Sustainability Professionals*.

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A TRACK RECORD THAT CONVERGED ON THIS

The Green Crypto Handbook

Rawson & Borreani, Taylor & Francis—the market thesis AqueductX operationalizes.

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Regen Atlas

1st place, Infrastructure & Digital Rights, PL_Genesis (572 submissions)—the registry and map AqueductX is built on.

→

SlabClaw

Agent-legibility proven on a live market—the same rails now pointed at commodity lots.

→

AqueductX

The fusion: handbook thesis, atlas registry, agent rails—one live build.

Ecofrontiers SARL—advises central banks and international organizations · ecofrontiers.xyz

$50k buys four deliverables. Public Good.

Open-source MIT—agents, intent pipeline, and a reference solver to fork.

01 **Harden three live verticals from existing partners on the Regen Atlas**
EthicHub (coffee trade finance), Glow (solar finance), LandX (agri-commodity trade finance).

02 **Wire the certification databases**
Resolve every lot against real certifier registries—Fairtrade, Rainforest Alliance, TIC—verified at the source, not declared.

03 **Sealed-bid solver races**
Solvers bid without seeing each other's prices, so coops and other producers keep bargaining power.

04 **Run a named-corridor pilot**
First cooperative or exporter reading its own lots in through the ERP integration—the point where the registry stops being our data and starts being theirs.

aqueductx.trade